

ABSTRACT

An anti-piracy system generates recipient-associated distinguishable versions of audio-video content. A reference version follows structured edit instructions identifying source cameras and camera-cut time codes. Mate versions retain ordered camera transitions at selected cuts but place the transitions at different timings, producing intervals in which reference and mate versions contain temporally corresponding frames from different cameras. Manifest files select recipient-associated chunk combinations representing reference-or-mate timing choices across the selected cuts, and a ledger associates delivered manifests with recipients. Camera-cut time codes detected in suspected unauthorized content produce a reconstructed manifest representing detected timing choices. The ledger identifies a recipient whose delivered manifest represents the same timing choice at each selected cut, including at least one mate timing. For a composite distribution, probabilistic fingerprinting compares detected timing choices for portions with recipient-associated sequences to identify multiple contributing recipients, and segmented Tardos processing localizes contributions by content segment.